

Impact of ambient temperature on parental well-being during pregnancy: a cohort study in Taiwan

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INTRODUCTION

- Climate change has emerged as one of the paramount global challenges in the 21st century. Previous studies indicate that rising ambient temperatures are associated with various adverse outcomes, including negative mood states (Noelke et al., 2016), increased hospitalizations due to mental and behavioral disorders (Dang et al., 2022), and even higher suicide rates (Kim et al., 2016).
- Evidence increasingly supports that certain groups are particularly vulnerable to the impacts of climate change. These groups include individuals with lower socioeconomic status, older adults, certain occupational groups, people with disabilities or chronic medical conditions, children, and pregnant women (USGCRP, 2016). **Research also suggests that extreme heat exposure may increase the risk of adverse pregnancy outcomes, such as preterm birth, low birth weight, stillbirth, and severe maternal morbidity** (Chersich et al., 2020; Jiao et al., 2023).
- However, research focusing on the impact of heat on mental health of pregnant populations, especially in Asia and island regions, remains limited.
- Therefore, this study aims to explore the relationship between ambient temperature and subjective well-being among expectant mothers and fathers.**

RESULTS

- The average age of mothers and fathers was 33.8 years (SD = 4.1) and 33.6 years (SD = 4.7), respectively. The sociodemographic profile of the participants is presented in Table 1, and the descriptive statistics of the exposure variables are shown in Table 2.
- Table 3 presents the results of the multiple linear regression models. After adjusting for potential confounders, our study found that **an increase in maximum temperature was associated with higher maternal eudaimonia and reduced negative affect during the first trimester** (eudaimonia: $\beta = 0.096$, 95% CI = [0.010, 0.183]; negative affect: $\beta = -0.063$, 95% CI = [-0.116, -0.011]). Additionally, **minimum temperature was associated with lower levels of negative affect** ($\beta = -0.036$, 95% CI = [-0.069, -0.004]) during the first trimester.
- The subgroup analysis revealed that **associations with maximum temperature were particularly significant for mothers expected to deliver in summer** (eudaimonia: $\beta = 0.109$, 95% CI = [0.007, 0.211]; negative affect: $\beta = -0.070$, 95% CI = [-0.132, -0.008]).
- However, these associations were not significant during the second or third trimesters. No significant relationships were observed between temperature and paternal well-being across all trimesters.

Table 1. Descriptive statistics of the study population. (n=577 pairs)

	Maternal	Paternal
	n (%)	n (%)
Education level		
High school and below	37 (6.4)	38 (6.7)
College	393 (68.4)	314 (55.1)
Graduate school	145 (25.2)	218 (38.2)
Household income		
< 60,000	92 (16.0)	—
60,000 ~ <100,000	271 (47.2)	—
≥ 100,000	211 (38.8)	—

Table 2. Descriptive statistics of exposure variables.

	First trimester	Second trimester	Third trimester
	Mean (SD)	Mean (SD)	Mean (SD)
Maximum temperature	29.5 (3.1)	3.1 (3.3)	21.3 (3.0)
Minimum temperature	28.9 (5.0)	3.3 (5.1)	22.2 (5.0)

Table 3. Maximum temperature and parental well-being.

	First trimester (n=575)		Second trimester (n=542)		Third trimester (n=533)	
	β	95% C.I.	β	95% C.I.	β	95% C.I.
Maternal						
Life satisfaction	0.034	(-0.029, 0.096)	0.028	(-0.027, 0.094)	-0.024	(-0.089, 0.041)
Eudaimonia	0.096	(0.010, 0.183)*	0.005	(-0.073, 0.084)	-0.051	(-0.148, 0.046)
Positive affect	0.010	(-0.040, 0.060)	0.005	(-0.039, 0.050)	0.012	(-0.040, 0.063)
Negative affect	-0.063	(-0.116, -0.011)*	0.005	(-0.041, 0.052)	0.018	(-0.037, 0.074)
Paternal						
Life satisfaction	0.032	(-0.035, 0.099)	0.014	(-0.049, 0.076)	0.023	(-0.048, 0.094)
Eudaimonia	0.051	(-0.040, 0.141)	0.046	(-0.046, 0.138)	0.031	(-0.070, 0.132)
Positive affect	-0.031	(-0.081, 0.018)	0.026	(-0.024, 0.077)	0.0002	(-0.049, 0.024)
Negative affect	-0.042	(-0.093, 0.009)	-0.007	(-0.056, 0.041)	-0.037	(-0.096, 0.021)

*p-value < 0.05, **p-value < 0.01, ***p-value < 0.001. Maximum temperature is presented as a continuous variable, with each unit increase representing one Celsius degree. Data analyzed using multiple linear regression. Results were adjusted for age, education level, and income.

METHODS

- Participants were from the **Longitudinal Examination across Prenatal and Postpartum Health in Taiwan (LEAPP-HIT)**, a prospective cohort study initiated in 2011 that enrolled expectant parents during early prenatal visits.
- A total of 577 pregnant pairs, collected between 2017 and 2022, were included.**
- Parental well-being was assessed using a self-reported 16-item scale adapted from the Organisation for Economic Cooperation and Development (OECD) Guidelines, with higher scores indicating greater life satisfaction, eudaimonia, positive affect, and worse levels of negative affect (OECD, 2013). Ambient temperature data were obtained from the Central Weather Administration of Taiwan.
- Multiple linear regression models were used for examination.

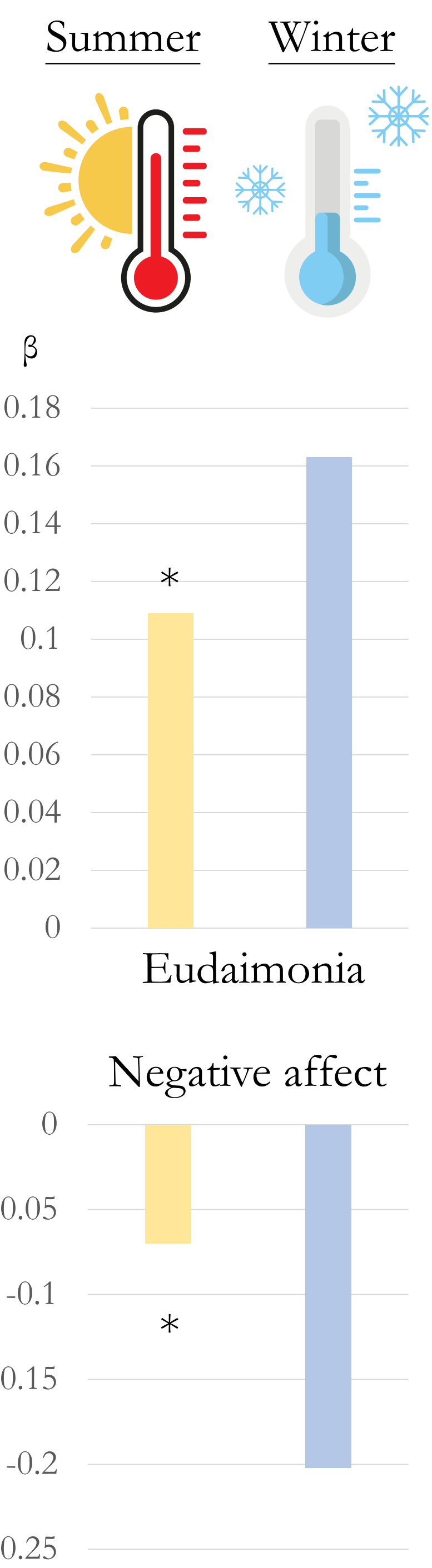


Figure 1. Subgroup analysis, maximum temperature and maternal well-being.

DISCUSSION

- Our findings suggest that, **especially for those delivered in summer, elevated maximum temperatures during the first trimester (typically during the colder months) were associated with improved maternal well-being in Taiwan.**
- Recent literature highlights that maternal positive affect may benefit fetal development, reducing the risk of preterm delivery (Voellmin et al., 2023). Moreover, positive maternal mental well-being may foster the brain development of the offspring (Qiu et al., 2024). Children of mothers with greater positive mental health during pregnancy are less likely to develop mental and behavioral disorders (Lähdepuro et al., 2023). Thus, promoting positive maternal mental health during pregnancy holds potential for improving the health of pregnant women and the developmental outcomes of future generations (Henrichs & Witteveen, 2023).
- This study adds to existing knowledge by highlighting the importance of **considering ambient temperature alongside individual and familial risk factors for maternal well-being during pregnancy**, as it impacts both the mother and child's health and development. Future research could explore the mechanisms underlying sex differences in the relationship between ambient temperature and mental health.

Strengths

- Inclusion of both mothers and fathers in the study
- Prospective cohort study design allows for comparison across different trimesters.
- Focus on positive mental health outcomes.
- Use of high-quality and precise ambient temperature data.

Limitations

- Self-reported mental health data may introduce bias.
- Unmeasured potential confounders could influence the results.
- Findings may not be generalizable to populations outside of urban Taiwan.

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